

Renormalization

1. Renormalization in ϕ^4 -theory

Firstly, we briefly recall some basic idea about renormalization. For counting the divergent of a theory, one may use the superficial degree of divergence:

$$D = \text{power of } k \text{ in numerator} - \text{power of } k \text{ in denominator.} \quad (2.1)$$

Suppose the Loop integrals in $d = 4$ space-time dimension have superficial degree of divergence D , which can be easily seen from the directly calculating:

$$\int^{\Lambda} d^4k \frac{k^{a-4}}{k^b} = \int^{\Lambda} k^3 dk d\Omega \frac{k^{a-b}}{k^4} \sim \int^{\Lambda} dk \frac{k^{a-b}}{k} = \begin{cases} \Lambda^D & \text{for } D = a - b \neq 0 \\ \ln \Lambda & \text{for } D = 0 \end{cases}, \quad (2.2)$$

where Λ is the cut-off. But we would like to use some more general form to think about it. As an example, considering the ϕ^n -theory

$$\mathcal{L} = \frac{1}{2}(\partial_{\mu}\phi)(\partial^{\mu}\phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{n!}\phi^n, \quad (2.3)$$

in which the propagators have the form of $(k^2 - m^2)^{-1}$, vertices are $\sim \lambda$ that connect n lines. So we can define So in this case, we can

1. $L = P - V + 1$

- L : loops (loop integral with d^4k)
- P : internal propagators (propagator with $(k^2)^{-1}$)
- V : vertices (vertices with energy-momentum conservation $\delta(\sum p_i)$)
- $+1$: global conservation of energy-momentum

2. $nV = 2P + N$

- n : number of lines at each vertex
- P : internal propagators
- N : external lines (or external propagator)

So for each loop, the power of k in numerator is 4, which is contributed by an integral measurement $\sim d^4k$, and the power of k in denominator is contributed by the propagator. Therefore, the superficial degree of divergence is

$$D = dL - 2P; \quad d = 4. \quad (2.4)$$

Thus, base on the relation we introduced above, we have

$$\begin{aligned} D &= 4(P - V + 1) - 2P \\ &= 2P - 4V + 4 \\ &= (n - 4)V - N + 4. \end{aligned} \quad (2.5)$$

Thus, we find D is independent of P , but only depends on N for the case $n = 4$. But for a more general n in $\frac{\lambda}{n!}\phi^n$, we have

- $n = 2$: $\implies$ free theory (included in the mass term);
- $n = 3$: $D = 4 - N - V \implies$ **super renormalizable**
- $n = 4$: $D = -N + 4 \implies$ **renormalizable**
- $n > 4$: $D = (n - 4)V - N + 4 \implies$ **non-renormalizable**

For a theory that the signature in front of V is “ $-$ ”, we call it as **super renormalizable theory** because the divergence only appear in some leading order terms, with respect the order getting higher, the diagram will finally be convergent, in which $D < 0$. *So in the super renormalizable theory only contains finite number of divergent degree*, which means only need finite counterterms. Similarly, for a theory that calls **renormalizable**, it is because it has no relation with V , which means *no matter how many orders we are counting, the divergent degrees are the same!* Although in this case, it contains infinite divergent diagrams, we can still by re-define the “physical” parameter to include the divergence. In the end, it is quite obvious to understand why we call the last situation as **non-renormalizable**. Because the pre-signature of V is positive, which means with the order getting higher, the divergent will be keep increasing, *so we need infinite parameter to cancel the divergence*, which in most of cases are technically impossible.

Another case is QED, we have two type of propagators, but they all contribute $\sim k^{-2}$

- N_γ : external photon lines;
- N_e : external electron lines;
- P_e : electron propagators;
- P_γ : photon propagators;
- $V = 2P_\gamma + N_\gamma = \frac{1}{2}(2P_e + N_e)$: number of vertices;
- $L = P_e + P_\gamma - V + 1$: number of the loops.

$$\begin{aligned}
 D &= 4L - P_e - 2P_\gamma \\
 &= 4(P_e + P_\gamma - V + 1) - P_e - 2P_\gamma \\
 &= 3P_e + 2P_\gamma - 4V + 4 \\
 &= 4 - N_\gamma - \frac{3}{2}N_e.
 \end{aligned} \tag{2.6}$$

So as we discussed above, since D is non-related with V , so QED is a renormalizable theory.

1.1 Analysis of dimensions in view of renormalization

Now we consider another way to analyze the renormalizability. We use the dimension of mass as a standard. So

$$[\text{Length}] = [\text{Time}] = [\text{Energy}]^{-1} = [\text{Mass}]^{-1}. \tag{2.7}$$

We require the dimension for the action $[S] = 0$, thus

$$[d^4x] = -4, \quad [\mathcal{L}] = 4, \quad [\partial_\mu] = 1, \quad [\phi] = 1. \tag{2.8}$$

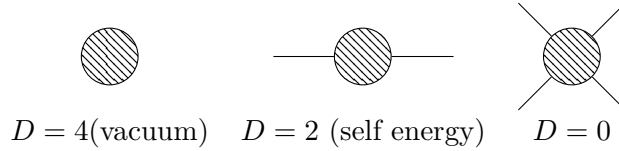
The last one is because $[\partial_\mu \phi \partial^\mu \phi] = 4$. The coupling constant $[\lambda]$ in $\frac{\lambda}{n!} \phi^n$ then has the dimension

$$[\lambda] = 4 - n. \quad (2.9)$$

Take a sharp look! This number, $4 - n$, is nothing but the prefactor of superficial degree of divergence in ϕ^n theory, i.e.,

$$D = 4 - N - \underbrace{(4 - n)}_{[\lambda]} V. \quad (2.10)$$

Thus, base on the discussion we did above, we find $[\lambda] < 0$ corresponds to a non-renormalizable theory. As an example, we consider the ϕ^4 -theory. So $n = 4$ brings the superficial degree of divergence is $D = 4 - N \geq 0$ for degree up to $N = 4$, which is a renormalizable theory.



$$D = 4(\text{vacuum}) \quad D = 2 \text{ (self energy)} \quad D = 0$$

We can also do the same thing for QED. For convenience, we rewrite the QED Lagrangian:

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \bar{\psi}(i\not{\partial} - m)\psi + e A_\mu \bar{\psi} \gamma^\mu \psi. \quad (2.11)$$

Similarly, since the action S is dimensionless, so

$$[\psi] = \frac{3}{2}, \quad [A_\mu] = 1. \quad (2.12)$$

Thus again, we can then see the coupling constant e is dimensionless, i.e., $[e] = 0$, which just corresponds to a renormalizable theory.

1.2 Renormalized perturbation theory

Now we still use ϕ^4 -theory as an example to show the renormalization procedure. The bare Lagrangian is

$$\mathcal{L}_{\text{bare}} = \frac{1}{2}(\partial_\mu \phi_0)(\partial^\mu \phi_0) - \frac{1}{2}m_0^2 \phi_0^2 + \frac{1}{4!} \lambda_0 \phi_0^4 \quad (2.13)$$

where the corner mark “0” represent the bare parameters, i.e., λ_0, m_0 , and ϕ_0 . This Lagrangian is nothing special, it just the original Lagrangian that we used in perturbation theory. Now we just give it a different name to separate with the renormalized Lagrangian that we will discuss later. The name “bare” represent this quantity is not actually the physical quantity that we observe because the bare parameters are actually infinity. These parameters appears only for cancel the infinity in the loop integral. The rest of this canceling is then the physical quantity, which is the renormalized quantity. This explanation shows the core idea of why do we need renormalization, i.e., the physical part of graphs needs to be observable, so we have to find a well-defined quantity that suitable for any order. But in this case, we may have to accept that many “constants” we thought unchanged will start to running with the higher energy scale being considered. And this will be described by the renormalization group equation.

The relation of the renormalized quantities with the bare quantities are

$$\phi_0 = \sqrt{Z} \phi_r, \quad Z m_0^2 = m^2 + \delta m^2, \quad Z^2 \lambda_0 = \lambda + \delta \lambda, \quad (2.14)$$

where $Z = 1 + \delta Z$. And δZ , δm^2 and $\delta\lambda$ are called the **counterterms**. So the bare Lagrangian can be written as

$$\begin{aligned}\mathcal{L}_{bare} &= \mathcal{L}_{ren} + \mathcal{L}_{count} \\ &= \underbrace{\frac{1}{2}(\partial_\mu\phi_r)(\partial^\mu\phi_r) - \frac{1}{2}m^2\phi_r^2 + \frac{\lambda}{4!}\phi_r^4}_{\mathcal{L}_{ren}} + \underbrace{\frac{1}{2}\delta Z(\partial_\mu\phi_r)(\partial^\mu\phi_r) - \frac{1}{2}\delta m^2\phi_r^2 + \frac{1}{4!}\delta\lambda\phi_r^4}_{\mathcal{L}_{counter\ term}}.\end{aligned}\quad (2.15)$$

The counterterms also have corresponding Feynman rules

$$\longrightarrow \otimes \longrightarrow = i(p^2\delta Z - \delta m^2); \quad \bigcirc = -i\delta\lambda. \quad (2.16)$$

And the Feynman rules of renormalized term is

$$\longrightarrow_p = \frac{i}{p^2 - m^2}; \quad \times = -mi\lambda. \quad (2.17)$$

The renormalization condition is usually called the **scheme**. The on-shell scheme for ϕ_r, m in two-point function is

$$\longrightarrow_p \text{ (with circle) } \longrightarrow_p = \frac{i}{p^2 - m_0^2 - \Sigma(p^2)} \quad (2.18)$$

$$= \frac{i}{p^2 - m_{phy}^2} + \underbrace{A + B(p^2 - m_{phy}^2) + C(p^2 - m_{phy}^2)^2 + \dots}_{\text{Regular Terms}}, \quad (2.19)$$

where the $\Sigma(p^2)$ is the self-energy correction. The main point of this scheme is requiring the residue at the pole $p^2 = m_{phys}^2$ to be normalized to 1. The mission of the counterterms is to absorb UV divergences and ensure the self-energy $\Sigma(p^2)$ satisfies the specific renormalization conditions, rather than cancelling the regular terms.

Now for the four-point functions, we impose a similar renormalization condition on the scattering amplitude:

$$\left(\text{diagram: two external lines meeting at a shaded circle} \right)_{\text{Amp}} = i\lambda, \quad (2.20)$$

at the threshold $s = 4m^2$ and $t = u = 0$, which corresponds to the case of zero momentum transfer. The Mandelstam variables are defined as usual: $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, and $u = (p_1 - p_4)^2$. This condition, together with the two-point function conditions mentioned earlier, defines the on-shell renormalization scheme. To implement this in practice, we follow these steps:

1. Calculate the divergent diagrams: Evaluate all Feynman diagrams contributing to the n -point functions at a given order in perturbation theory. A suitable regularization scheme (e.g., dimensional regularization) must be chosen to handle the UV divergences.
2. Determine the counterterms: Extract the specific values of the counterterms δm^2 , δZ , and $\delta\lambda$ by requiring that the renormalized Green's functions satisfy the chosen renormalization conditions (such as the one in (2.20)).

To conclude this section, we will briefly do the renormalization process for the 1-loop for ϕ^4 -theory. Firstly, we expand the two point function until λ^2 :

$$\text{---} \text{---} \text{---} \text{---} \text{---} = \text{---} \text{---} \text{---} + \left(\text{---} \text{---} \text{---} + \text{---} \otimes \text{---} \right) + O(\lambda^2), \quad (2.21)$$

where the first term is just the normal propagator, the second and the third is the 1-loop graph and the counterterm. Thus, we have

$$\text{---} \text{---} \text{---} \text{---} \text{---} = \frac{i}{p^2 - m^2} - \frac{i}{2} \lambda \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} + i(p^2 \delta Z - \delta m^2) + O(\lambda^2). \quad (2.22)$$

We can by directly calculating the tadpole integral to separate the divergent part, such that to determine the expression for the counterterms. As a result, we only shows the final expression that you should get:

$$\delta Z = 0 \quad (2.23)$$

$$\delta m^2 = -\frac{\lambda}{2} \frac{\Gamma(1 - d/2)}{(4\pi)^{d/2} (m^2)^{1-d/2}}. \quad (2.24)$$

If we take $d = 4 - 2\epsilon$, the theory is renormalizable, so the form of the counterterms are suitable for any order.

The 4-point function also can be expanded into 1-loop order, but since there are two vertices, so the next to next leading order we have is λ^3 :

$$\text{---} \otimes \text{---} = \text{---} \times \text{---} + \left(\text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} + \text{---} \text{---} \text{---} \right) + \text{---} \otimes \text{---} + O(\lambda^3) \quad (2.25)$$

$$= -i\lambda + (-i\lambda)^2 (iV(s) + iV(t) + iV(u)) - i\delta\lambda + O(\lambda^3) \quad (2.26)$$

The counterterm $\delta\lambda$ is determined by requiring the total scattering amplitude to satisfy the renormalization condition at the chosen subtraction point. According to the Feynman rules, the total amplitude at 1-loop order is the sum of the tree-level vertex, the loop corrections (s, t, and u channels), and the counterterm vertex. For each $iV(p^2)$, we have

$$iV(p^2) = \frac{i}{2} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2} \frac{i}{(k+p)^2 - m^2} \quad (2.27)$$

$$= -\frac{i}{2} \frac{1}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma(2 - d/2)}{(m^2 - x(1-x)p^2)^{2-d/2}}, \quad (2.28)$$

where x is the Feynman parameter. By taking $d = 4 - 2\epsilon$, we can separate the divergent part too:

$$iV(p^2) = -\frac{i}{32\pi^2} \int_0^1 dx \left(\frac{1}{\epsilon} - \gamma_E + \ln(4\pi) - \ln(m^2 - x(1-x)p^2) + O(\epsilon) \right). \quad (2.29)$$

To satisfy the on-shell condition defined in Eq. (1.20), we require the amplitude to be exactly $-i\lambda$ at the threshold $s = 4m^2, t = u = 0$. This implies that the loop contributions and the counterterm must cancel each other out at this specific kinematic point:

$$(-i\lambda)^2 [iV(4m^2) + iV(0) + iV(0)] - i\delta\lambda = 0 \quad (2.30)$$

Solving for $\delta\lambda$ yields the expression for the 1-loop counterterm in ϕ^4 theory:

$$\delta\lambda = -\lambda^2 [V(4m^2) + 2V(0)], \quad (2.31)$$

Therefore, the divergence $1/\epsilon$ pole contained in $V(p^2)$ is exactly absorbed by $\delta\lambda$, rendering the final physical amplitude finite. This ensures that the coupling constant λ corresponds to the strength of the interaction measured at the specific energy scale $s = 4m^2$.

2. Wilson's approach to renormalization

The main idea of Wilson's renormalization is introducing a cut-off Λ and compute how the theory changes at different length scales. For that, we need to working with the generating functional with cut-off and successive integration over large momenta/small length scales:

$$Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]}, \quad (2.32)$$

where we used Wick rotation changing to Euclidean space-time:

$$x^0 \rightarrow ix^4, \quad d^4x \rightarrow id^4x_E. \quad (2.33)$$

Thus, the measurement is

$$\mathcal{D}\phi_\Lambda = \prod_{|k| < \Lambda} d\phi(k). \quad (2.34)$$

In this case, the modes of ϕ are well defined in Euclidean space-time such that we can make sure that when $k \rightarrow \infty$ all the components of momentum also go to infinity. This step is to avoid in Minkowski space-time $k^2 = (k^0)^2 - (k^i)^2$. So when $k^2 \rightarrow \infty$, it could be $k^0 \rightarrow \infty$ and $k^i \rightarrow 0$, which doesn't suitable for our analysis.

Now, the action and Lagrangian become

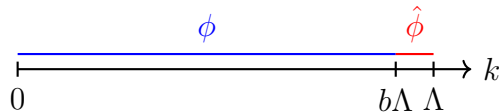
$$S[\phi] = \int d^4x \mathcal{L}, \quad (2.35)$$

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4. \quad (2.36)$$

Suppose the mass $m \ll \Lambda$, and the coupling constant $\lambda \ll 1$, which means the last two terms in the Lagrangian can be treated as perturbation. For going to the perturbation theory, it's useful to split up fields $\phi \rightarrow \phi + \hat{\phi}$ as

$$\hat{\phi} = \begin{cases} \phi(k) & \text{for } b\Lambda \leq |k| \leq \Lambda \\ 0 & \text{for } |k| \leq b\Lambda \end{cases}; \quad \phi = \begin{cases} 0 & \text{for } b\Lambda \leq |k| \leq \Lambda \\ \phi(k) & \text{for } |k| \leq b\Lambda \end{cases}, \quad (2.37)$$

where $(b\Lambda, \Lambda)$ is a very small interval controlled by parameter $b \leq 1$



Then we can also split up the action as $S[\phi] \rightarrow S[\phi] + \hat{S}[\phi, \hat{\phi}]$, so we have

$$S[\phi] = \int d^4x \frac{1}{2} \partial_\mu (\phi + \hat{\phi}) \partial^\mu (\phi + \hat{\phi}) + \frac{1}{2} m^2 (\phi + \hat{\phi})^2 + \frac{\lambda}{4!} (\phi + \hat{\phi})^4 \quad (2.38)$$

$$= \int d^4x \frac{1}{2} \left(\partial_\mu \phi \partial^\mu \phi + 2 \partial_\mu \phi \partial^\mu \hat{\phi} + \partial_\mu \hat{\phi} \partial^\mu \hat{\phi} \right) + \frac{1}{2} m^2 (\phi^2 + \hat{\phi}^2 + 2\phi\hat{\phi}) \quad (2.39)$$

$$+ \frac{\lambda}{4!} \left(\phi^4 + \phi^2 \hat{\phi}^2 + 2\phi^3 \hat{\phi} + \hat{\phi}^2 \phi^2 + \hat{\phi}^4 + 2\phi \hat{\phi}^3 + 2\phi^3 \hat{\phi} + 2\phi \hat{\phi}^3 + 4\phi^2 \hat{\phi}^2 \right) \quad (2.40)$$

$$= \int d^4x \left[\underbrace{\frac{1}{2} (\partial_\mu \phi)^2 + \frac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!} \phi^4}_{\mathcal{L}[\phi]} + \underbrace{\frac{1}{2} (\partial_\mu \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 + \frac{\lambda}{4!} (6\phi^2 \hat{\phi}^2 + 4\phi^3 \hat{\phi} + 4\phi \hat{\phi}^3 + \hat{\phi}^4)}_{\hat{\mathcal{L}}[\phi, \hat{\phi}]} \right], \quad (2.41)$$

where we have dropped terms containing the products $\phi \hat{\phi}$ because Fourier components are orthogonal:

$$\int \frac{d^4k}{(2\pi)^4} \frac{d^4k'}{(2\pi)^4} \phi(k) \hat{\phi}(k') e^{i(k+k') \cdot x} \sim \delta^{(4)}(k+k') = 0 \quad (2.42)$$

Thus, we have

$$\hat{S}[\phi, \hat{\phi}] = \int d^4x (\hat{\mathcal{L}}_0[\hat{\phi}] + \hat{\mathcal{L}}_{\text{int}}[\phi, \hat{\phi}]), \quad (2.43)$$

and

$$\hat{\mathcal{L}}_0[\hat{\phi}] = \frac{1}{2} \partial_\mu \hat{\phi} \partial^\mu \hat{\phi}, \quad (2.44)$$

$$\hat{\mathcal{L}}_{\text{int}}[\phi, \hat{\phi}] = \frac{1}{2} m^2 \hat{\phi}^2 + \lambda \left(\frac{1}{3!} \phi \hat{\phi}^3 + \frac{1}{4} \phi^2 \hat{\phi}^2 + \frac{1}{3!} \phi^3 \hat{\phi} + \frac{1}{4!} \hat{\phi}^4 \right). \quad (2.45)$$

Thus, the first part in $\hat{S}$ can be rewritten by the Fourier transformation as

$$\int d^4x \hat{\mathcal{L}}_0[\hat{\phi}] = \int \frac{d^4k d^4p}{(2\pi)^8} (ik_\mu \hat{\phi}(k)) (ip^\mu \hat{\phi}(p)) \int d^4x e^{i(k+p) \cdot x} \quad (2.46)$$

$$= \int \frac{d^4k d^4p}{(2\pi)^8} (ik_\mu)(ip^\mu) \hat{\phi}(k) \hat{\phi}(p) (2\pi)^4 \delta^{(4)}(k+p) \quad (2.47)$$

$$= \frac{1}{2} \int_{b\Lambda < |k| < \Lambda} \frac{d^4k}{(2\pi)^4} \hat{\phi}(-k) k^2 \hat{\phi}(k) \quad (2.48)$$

$$= \frac{1}{2} \int_{b\Lambda < |k| < \Lambda} \frac{d^4k}{(2\pi)^4} \hat{\phi}^*(k) k^2 \hat{\phi}(k). \quad (2.49)$$

This Gaussian integral form remind us to write a propagator for $\hat{\phi}$ from Wick contraction of $\hat{\phi}$ fields:

$$\overline{\hat{\phi}(k) \hat{\phi}(p)} = \langle \hat{\phi}(k) \hat{\phi}(p) \rangle = \frac{\int \mathcal{D}\hat{\phi} \hat{\phi}(k) \hat{\phi}(p) e^{-S_0[\hat{\phi}]}}{\int \mathcal{D}\hat{\phi} e^{-S_0[\hat{\phi}]}} = \frac{1}{k^2} (2\pi)^4 \delta^{(4)}(k+p) \Theta(k), \quad (2.50)$$

where

$$\Theta(k) = \begin{cases} 1 & \text{for } b\Lambda < |k| < \Lambda \\ 0 & \text{else} \end{cases}. \quad (2.51)$$

Therefore, we can separate the generating functional measurement with the **high energy modes** $\hat{\phi}$ and **low energy modes** $\phi_{b\Lambda}$ as $\mathcal{D}\phi_\Lambda = \mathcal{D}\phi_{b\Lambda}\mathcal{D}\hat{\phi}_\Lambda$. And we can integrate the high energy modes

$$Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]} = \int \mathcal{D}\phi_{b\Lambda} \mathcal{D}\hat{\phi}_\Lambda e^{-S[\phi] - \hat{S}[\phi, \hat{\phi}]} = \int \mathcal{D}\phi_{b\Lambda} e^{-S[\phi]} \left(\int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi, \hat{\phi}]} \right) \quad (2.52)$$

$$= \int \mathcal{D}\phi_{b\Lambda} e^{-S_{\text{eff}}[\phi]}, \quad (2.53)$$

where left a **effective action** defined as

$$S_{\text{eff}}[\phi] = S[\phi] + \delta S[\phi], \quad (2.54)$$

in which $\delta S[\phi]$ can be treated as perturbation, thus it can be expanded as

$$e^{-\delta S[\phi]} = \int \mathcal{D}\hat{\phi}_\Lambda e^{-\hat{S}[\phi, \hat{\phi}]} = \int \mathcal{D}\hat{\phi}_\Lambda e^{-\int d^4x \hat{\mathcal{L}}[\phi, \hat{\phi}]} \quad (2.55)$$

$$\simeq \int \mathcal{D}\hat{\phi}_\Lambda \int d^4x \left[1 - \left(\hat{\mathcal{L}}_0 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{3!}\phi\hat{\phi}^3 + \frac{\lambda}{4}\phi^2\hat{\phi}^2 + \frac{\lambda}{3!}\phi^3\hat{\phi} + \frac{\lambda}{4!}\hat{\phi}^4 \right) + \mathcal{O}(\lambda^2) \right] \quad (2.56)$$

$$= \int \mathcal{D}\hat{\phi}_\Lambda \int d^4x \left[1 - \underbrace{\left(\hat{\mathcal{L}}_0 + \frac{1}{2}m^2\hat{\phi}^2 + \frac{\lambda}{4!}\hat{\phi}^4 \right)}_{\substack{\text{Doesn't contain } \phi, \\ \text{which has no contribution for running}}} - \frac{\lambda}{4}\phi^2\hat{\phi}^2 + \mathcal{O}(\lambda^2) \right], \quad (2.57)$$

where in the third step we have dropped all the odd terms which don't contribute to the integral. The introducing of $\delta S[\phi]$ will change the parameters in the original action $S[\phi]$, which will finally leading the parameters start running. To show this, we now need to work in the perturbation scenario. Since we have assumed that $m \ll \Lambda$ and $\lambda \ll 1$, so we can expand $e^{-S_{\text{eff}}[\phi]}$ with Dyson series in terms of λ . For that, we can generally write the effective Lagrangian $\mathcal{L}_{\text{eff}}$ for effective action $S_{\text{eff}}[\phi]$ as:

$$\mathcal{L}_{\text{eff}}[\phi] = \frac{1}{2}(1 + \Delta Z)\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}(m^2 + \Delta m^2)\phi^2 + \frac{1}{4!}(\lambda + \Delta\lambda)\phi^4 + (\text{higher dimensional operators}). \quad (2.58)$$

Now the mission is to determine ΔZ , Δm^2 and $\Delta\lambda$. We firstly consider the 1-loop order $\mathcal{O}(\lambda)$ correction for the propagator, which corresponds to the term $\frac{1}{4}\lambda\phi^2\hat{\phi}^2$, the graph is :

$$I_1 = \phi \text{ --- } \text{ (double line loop) } \text{ --- } \phi, \quad \mathcal{O}(\lambda) \quad (2.59)$$

where the double line represent the contraction of $\hat{\phi}$, which doesn't contribute the real physical observable, thus it needs to be integrated. Thus, we have

$$I_1 = -\frac{\lambda}{4} \int d^4x \phi^2 \hat{\phi} \hat{\phi} \quad (2.60)$$

$$= -\frac{\lambda}{4} \int d^4x \int \frac{d^4k_1}{(2\pi)^4} \dots \frac{d^4k_4}{(2\pi)^4} e^{i(k_1 + \dots + k_4) \cdot x} \phi(k_1) \phi(k_2) \frac{1}{k_3^2} (2\pi)^4 \delta^{(4)}(k_3 + k_4) \Theta(k_3), \quad (2.61)$$

integrate for x, k_4 :

$$I_1 = -\frac{\lambda}{4} \int \frac{d^4 k_1}{(2\pi)^4} \cdots \frac{d^4 k_3}{(2\pi)^4} \phi(k_1) \phi(k_2) \frac{1}{k_3^2} (2\pi)^4 \delta^{(4)}(k_1 + k_2) \Theta(k_3) \quad (2.62)$$

$$= -\frac{1}{2} \mu \int \frac{d^4 k_1}{(2\pi)^4} \phi(k_1) \phi(-k_1), \quad (2.63)$$

where

$$\mu = \frac{\lambda}{2} \int_{b\Lambda < |k| < \Lambda} \frac{d^4 k}{(2\pi)^4} \frac{\Theta(k)}{k^2} \quad (2.64)$$

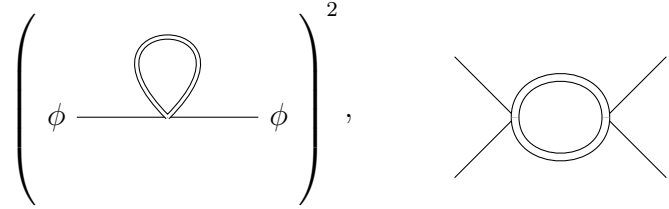
$$= \frac{\lambda}{2} \frac{1}{(2\pi)^4} \int d\Omega \int_{b\Lambda}^{\Lambda} k dk \quad (2.65)$$

$$= \frac{\lambda}{32\pi^2} (1 - b^2) \Lambda^2. \quad (2.66)$$

I_1 has structure of mass correction, which can be put in exponent of effective action as

$$\exp \left\{ - \int d^4 x \frac{1}{2} \mu \phi^2 \right\}. \quad (2.67)$$

And the $\mathcal{O}(\lambda^2)$ order graphs therefore are:



For the first one, it's nothing but the disconnected graph $\sim \frac{1}{2} I_1^2$, which contribute $\mathcal{O}(\lambda^2)$ in the exponent. The second graph represent the correction for the ϕ^4 interaction vertex:

$$I_2 = -\frac{1}{4!} \xi \int d^4 x \phi^4(x), \quad (2.68)$$

factorizing the loop integral, then we have

$$\xi = -4! \cdot \underbrace{2}_{\substack{\text{2 possible} \\ \text{contractions} \\ \text{of } \phi^2(\phi\phi)^2}} \cdot \underbrace{\frac{1}{2!}}_{\substack{\text{symmetry factor} \\ \text{of diagram}}} \cdot \left(\frac{\lambda}{4} \right)^2 \int_{b\Lambda < |k| < \Lambda} \frac{d^4 k}{(2\pi)^4} \left(\frac{1}{k^2} \right)^2 \Theta(k) = -\frac{3\lambda^2}{16\pi^2} \ln \frac{1}{b} \quad (2.69)$$

Thus, we find μ, ξ can be interpreted as combinations to $\Delta m^2, \Delta\lambda$.

Therefore, the generating functional $Z = \int \mathcal{D}\phi_\Lambda e^{-S[\phi]}$ changing to $Z = \int \mathcal{D}\phi_{b\Lambda} e^{-S_{\text{eff}}[\phi]}$ is by rescaling S to effective action S_{eff} by changing the coordinates and momenta:

$$k' = \frac{k}{b}, \quad x' = bx \quad (2.70)$$

Thus,

$$S_{\text{eff}} = \int d^4 x' \frac{1}{b^4} \left(\frac{1}{2} (1 + \Delta Z) b^2 \partial'_\mu \phi \partial'^\mu \phi + \frac{1}{2} (m^2 + \Delta m^2) \phi^2 + \frac{1}{4!} (\lambda + \Delta\lambda) \phi^4 + \underbrace{\cdots + \frac{1}{n!} (\lambda_n + \Delta\lambda_n) \phi^n + \cdots}_{\text{higher dim. operators}} \right) \quad (2.71)$$

$$= \int d^4 x' \left(\frac{1}{2} \partial'_\mu \phi' \partial'^\mu \phi' + \frac{1}{2} m'^2 \phi'^2 + \frac{1}{4!} \lambda' \phi'^4 + \cdots + \frac{1}{n!} \lambda'_n (\phi')^n + \cdots \right), \quad (2.72)$$

where

$$\phi' = \sqrt{(1 + \Delta Z)b^{-2}}\phi \quad (2.73)$$

$$m'^2 = (m^2 + \Delta m^2)(1 + \Delta Z)^{-1}b^{-2} \quad (2.74)$$

$$\lambda' = (\lambda + \Delta\lambda)(1 + \Delta Z)^{-2}b^0 \quad (2.75)$$

$$\lambda'_n = (\lambda_n + \Delta\lambda_n)(1 + \Delta Z)^{-n/2}b^{n-4}. \quad (2.76)$$

This process, mapping $\phi' \rightarrow \phi$, $m'^2 \rightarrow m^2$, $\lambda' \rightarrow \lambda$, is exactly the **renormalization group transformations**. We can also explicitly show the relations for rescaling of parameters for mapping $x' = bx$ on $D = 4$ dimension:

$$m'^2 = b^{-2}\Delta m^2 \quad (2.77)$$

$$\lambda' = b^0\Delta\lambda \quad (2.78)$$

$$\lambda'_n = b^{n-4}\Delta\lambda_n \quad (2.79)$$

Generally, in D dimensions, we have:

$$m'^2 = b^{-2}\Delta m^2 \quad (2.80)$$

$$\lambda' = b^{D-4}\Delta\lambda \quad (2.81)$$

$$\lambda'_n = b^{D-4+n-4}\Delta\lambda_n \quad (2.82)$$

for $b < 1$ on $D = 4$:

- Super-renormalizable m^2 grows as $b^{-2} \longleftrightarrow$ relevant operator ϕ^2 .
- Renormalizable λ scales as $b^0 \longleftrightarrow$ marginal operators $\lambda\phi^4$.
- Non-renormalizable λ_n vanishes as $b^{n-4} \longleftrightarrow$ irrelevant operators $\lambda_n\phi^n, n > 4$.

$$\lambda' = \lambda - \frac{3}{16\pi^2}\lambda^2 \ln\left(\frac{1}{b}\right) + O(\lambda^3) \quad (2.83)$$

3. Callan-Symanzik Equation

Goal: Quantify scale dependence of Green's functions $G^{(n)}(x_1 \dots x_n, M)$, where M sets the scale of external momenta, which leads to the Callan-Symanzik equation.

3.1 Alternative choice for renormalization in ϕ^4 -theory

In ϕ^4 -theory, the 1-PI graphs of propagator can be renormalized by taking the renormalization scheme. We simply recall this process again, firstly, taking

$$\text{---} \bigcirc \text{1PI} \text{---} = -i\Sigma(p^2), \quad (2.84)$$

for all 1-PI contributions to propagators. Thus, the full propagator goes like:

$$\begin{aligned} \text{---} \text{---} \bigcirc \text{---} \text{---} &= \text{---} \text{---} \text{---} + \text{---} \text{---} \bigcirc \text{1PI} \text{---} \text{---} + \text{---} \text{---} \bigcirc \text{1PI} \text{---} \bigcirc \text{1PI} \text{---} + \dots \\ &= \frac{i}{p^2 - m^2 - \Sigma(p^2)}. \end{aligned} \quad (2.85)$$

Here we took the on-shell scheme:

$$\Sigma(p^2)|_{p^2=m_{ph}^2} = 0, \quad \frac{d}{dp^2}\Sigma(p^2)|_{p^2=m_{ph}^2} = 0. \quad (2.86)$$

This scheme force the residue around the pole to be unity. Then we obtained the renormalized propagator:

$$\begin{array}{c} \xrightarrow{p} \\ \text{---} \circ \text{---} \xrightarrow{p} \end{array} = \frac{i}{p^2 - m_{ph}^2}. \quad (2.87)$$

But this is not physical in many cases, say QCD and SM due to many of problems. For example, since the quarks are confined so normally we can hardly define the asymptotic state to imagine a free quark. And the massless field of SM in this scheme will meet IR-divergence, which will also lead us to many unnecessary problems. So we need to choose other better schemes for a more general model. One alternative is to define counterterms at arbitrary scale M , at the external space-like momenta $p^2 = -M^2 < 0$, we take

$$\left(\begin{array}{c} \xrightarrow{p} \\ \text{---} \circlearrowright \text{1PI} \text{---} \\ \xrightarrow{p} \end{array} \right)_{p^2=-M^2} = 0 \quad , \quad \frac{\mathrm{d}}{\mathrm{d} p^2} \left(\begin{array}{c} \xrightarrow{p} \\ \text{---} \circlearrowright \text{1PI} \text{---} \\ \xrightarrow{p} \end{array} \right)_{p^2=-M^2} = 0,$$

$$\left(\begin{array}{c} p_1 \\ \swarrow \quad \nearrow \\ \bullet \\ \nwarrow \quad \searrow \\ p_2 \quad p_3 \end{array} \right)_{s=t=u=-M^2} = i\lambda. \tag{2.88}$$

where $(p_1 + p_2)^2 = s$, $(p_1 + p_3)^2 = t$, $(p_1 + p_4)^2 = u$. The reason why we choose the space-like momentum is to avoid poles. If we choose a time-like momentum, it for sure corresponds to physical particle, however also having divergence. The first condition $\Sigma(-M^2) = 0$ requires the correction for mass Σ_{finite} is 0 at this non-fixing scale M , and all the divergence Σ_{div} will be subtracted by the counterterms. So basically it looks very similar with the on-shell condition, the only thing that we change is changing the fixed physical mass m^2 to the unphysical mass scale $-M^2$.

This arbitrary parameter M is called the **renormalization scaling**. These conditions defined the only certain value at a fixed point, which canceled all the divergents. So we say **it defines the theory in scale M** .

For seeing this scheme is indeed valid, we firstly look at the Green's function with renormalized fields $\phi = \sqrt{Z}\phi_r$ and coupling λ , and taking the renormalized scale as M , we have

$$\begin{aligned}
G_{\text{ren}}^{(n)}(x_1 \dots x_n, \lambda, M) &= \langle \Omega | T \{ \phi_r(x_1) \dots \phi_r(x_n) \} | \Omega \rangle \\
&= Z^{-n/2} \langle \Omega | T \{ \phi(x_1) \dots \phi(x_n) \} | \Omega \rangle \\
&= Z^{-n/2} \underbrace{G_{\text{bare}}^{(n)}(x_1 \dots x_n, \lambda, M)}_{\text{bare Green's fun. independent of } M} .
\end{aligned} \tag{2.89}$$

Now we change variables as $M \rightarrow M + \delta M$. This changing of our scaling will also change the other parameters to satisfy the renormalization condition. Thus it leads to $\lambda \rightarrow \lambda + \delta\lambda$ and $\phi_{\text{ren}} \rightarrow \phi'_{\text{ren}} = (1 + \delta\eta)\phi_{\text{ren}}$, thus

$$(Z')^{-1/2}\phi_0 = Z^{-1/2}(1 + \delta eta)\phi_0 \implies Z' = Z(1 + \delta\eta)^{-2} \simeq Z(1 - 2\delta\eta) \equiv Z(1 + \delta Z), \quad (2.90)$$

where $\delta\eta = \frac{-\delta Z}{2}$. The latter gives the way of Green's function changing:

$$G_{\text{ren}}^{(n)} \rightarrow Z^{-n/2}(1 + \delta Z)^{-n/2} G_{\text{bare}}^{(n)} \quad (2.91)$$

$$\simeq Z^{-n/2}(1 - \frac{n}{2}\delta Z) G_{\text{bare}}^{(n)} \quad (2.92)$$

$$= Z^{-n/2}(1 + n\delta\eta) G_{\text{bare}}^{(n)} \quad (2.93)$$

$$= (1 + n\delta\eta) G_{\text{ren}}^{(n)}, \quad (2.94)$$

We will ignore the corner index “ren” in $G_{\text{ren}}^{(n)}$, **namely the renormalized Green's function denotes as $G^{(n)}$** . Thus from $\delta G_{\text{bare}} = 0$, we have

$$0 = \delta(Z^{n/2} G^{(n)}) = \underbrace{(\delta Z^{n/2}) G^{(n)}}_{n\delta\eta Z^{n/2}} + Z^{n/2} \frac{\partial}{\partial M} G^{(n)} \delta M + Z^{n/2} \frac{\partial}{\partial \lambda} G^{(n)} \delta \lambda. \quad (2.95)$$

Introducing the new variables:

$$\beta = \frac{M}{\delta M} \delta \lambda, \quad \gamma = \frac{M}{\delta M} \delta \eta \quad (2.96)$$

Thus, dropping the arbitrary δM , we obtain **the Callan-Symanzik equation**:

$$\left(M \frac{\partial}{\partial M} + \beta \frac{\partial}{\partial \lambda} + n\gamma \right) G^{(n)}(x_1 \dots x_n, \lambda, M) = 0, \quad (2.97)$$

where β and γ are universal for a given theory, which is independent for particular Green's function. This also means that these parameters are independent of any UV-cut off (dimensionless).

$$\beta = \beta(\lambda), \quad \gamma = \gamma(\lambda). \quad (2.98)$$

We can also directly compute $dG^{(n)}$ as

$$dG^{(n)} = (1 + n\delta\eta) G^{(n)} - G^{(n)} = n\delta\eta G^{(n)} = n \frac{dM}{M} \left(M \frac{d\eta}{dM} \right) G^{(n)} \quad (2.99)$$

$$= \frac{\partial G^{(n)}}{\partial M} dM + \frac{\partial G^{(n)}}{\partial \lambda} d\lambda \quad (2.100)$$

$$= \frac{\partial G^{(n)}}{\partial M} dM + \left(M \frac{d\lambda}{dM} \right) \frac{dM}{M} \frac{\partial G^{(n)}}{\partial \lambda} \quad (2.101)$$

$$= \frac{dM}{M} \left[M \frac{\partial}{\partial M} + \left(M \frac{d\lambda}{dM} \right) \frac{\partial}{\partial \lambda} \right] G^{(n)}, \quad (2.102)$$

from this construction, we can also obtain the Callan-Symanzik equation.

3.2 β -function in massless ϕ^4 -theory

Now let's focus some actual cases to find the *beta*-function and the anomalous dimension $\gamma(\lambda)$. For simplicity, we consider the 1-loop correction for **massless ϕ^4 -theory**. The Callan-Symanzik equation for $G^{(2)}$ gives us $\gamma = 0 + O(\lambda^2)$, because in 1-loop order, there's no

contribution to Z , but only to δm^2 . Only until 2-loops or higher orders which contribute Z and also of course δm^2 .

$$\begin{aligned}
 \text{---} \bigcirc \text{1PI} \text{---} &= \text{---} + \text{---} \bigcirc \text{---} + \frac{1}{2} \left(\text{---} \bigcirc \text{---} \right)^2 + \text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} + O(\lambda^3). \\
 &\quad (2.103)
 \end{aligned}$$

For $\beta(\lambda)$ at $O(\lambda)$, we need to consider the 4-point function correction:

$$G^{(4)}(p) = \text{---} \times \text{---} + \left(\text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} + \text{---} \bigcirc \text{---} \right) + \text{---} \bigcirc \text{---} + O(\lambda^3) \quad (2.104)$$

$$= -i\lambda + (-i\lambda)^2(iV(t) + iV(s) + iV(u)) - i\delta\lambda + O(\lambda^3), \quad (2.105)$$

where the renormalization condition are taken in $t = s = u = -M^2$. So for each diagram vertex correction, for $m = 0$ we have:

$$V(p^2) = \frac{1}{2} \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2} \frac{i}{(p-k)^2}. \quad (2.106)$$

Counterterm $\delta\lambda$ determined from amputated Green's function

$$G_{\text{amp}}^{(4)}|_{s=t=u=-M^2} \rightarrow \delta\lambda = (-i\lambda)^2 3V(-M^2). \quad (2.107)$$

Take $D = 4 - 2\epsilon$:

$$V(-M^2) = \lim_{\epsilon \rightarrow 0} \frac{1}{2} \frac{-1}{(4\pi)^{d/2}} \int_0^1 dx \frac{\Gamma(\epsilon)}{(x(1-x)M^2)^\epsilon} \quad (2.108)$$

$$= -\frac{1}{32\pi^2} \left(\frac{1}{\epsilon} - \ln M^2 + O(1) \right). \quad (2.109)$$

Thus

$$\delta\lambda = \frac{3\lambda^2}{32\pi^2} \left(\frac{1}{\epsilon} - \ln M^2 + O(1) \right). \quad (2.110)$$

This will lead to

$$G^{(4)} = -i\lambda + 3i(i\lambda)^2 V(p^2) - i\delta\lambda + O(\lambda^3) \quad (2.111)$$

$$= -i\lambda + \frac{3i\lambda^2}{32\pi^2} \left(\frac{1}{\epsilon} - \ln p^2 - \frac{1}{\epsilon} + \ln M^2 \right) + O(\lambda^3) \quad (2.112)$$

$$= -i\lambda + \frac{3i\lambda^2}{32\pi^2} \ln \left(\frac{M^2}{p^2} \right) + O(\lambda^3) \quad (2.113)$$

Thus, we have

$$M \frac{\partial}{\partial M} G^{(4)} = \frac{3i\lambda^2}{16\pi^2} + O(\lambda^3). \quad (2.114)$$

Thus, bring it into the Callan-Symanzik equation, we have

$$-\beta(\lambda) \frac{\partial}{\partial \lambda} G^{(4)} = i \frac{3\lambda^2}{16\pi^2} + O(\lambda^3) \quad (2.115)$$

where $\gamma = 0 + O(\lambda^2)$. Since we only consider order $O(\lambda^2)$, therefore we only need to expand $G^{(4)}$ into $O(\lambda)$, i.e.,

$$\frac{\partial}{\partial \lambda} G^{(4)} = -i + \frac{3i\lambda}{16\pi^2} \ln \left(\frac{M^2}{p^2} \right) + O(\lambda^2) \simeq -i + O(\lambda), \quad (2.116)$$

thus the β -function at 1-loop is:

$$\beta(\lambda) = \frac{3}{16\pi^2} \lambda^2 + O(\lambda^3). \quad (2.117)$$

3.3 Analogous analysis for massless QED

Now we consider the QED, which as a good example to calculate the anomalous dimension $\gamma(\lambda)$. In QED, we need to renormalize the vertex(Z_1) and the propagator for electron(Z_2) and photon(Z_3). These three parameters defined the electron charge renormalization:

$$e_0 = Z_1^{-1} Z_2 Z_3^{1/2} e. \quad (2.118)$$

The Green's function $G^{(n_e, n_A)}(x_1 \dots x_n, e, M)$ now contains two fields, so the Callan-Symanzik equation is

$$\left(M \frac{\partial}{\partial M} + \beta(e) \frac{\partial}{\partial e} + n_e \gamma_2 + n_A \gamma_3 \right) G^{(n_e, n_A)}(x_1 \dots x_n, e, M) = 0, \quad (2.119)$$

where n_e denote for the electron fields, n_A for photon fields. And γ_2, γ_3 are anomalous dimensions of electron and photon field respectively

$$\gamma_i = \frac{1}{2} M \frac{\partial}{\partial M} \delta_i, \quad i = 2, 3. \quad (2.120)$$

The γ_i comes from the field renormalization, which related with Z_2 and Z_3 . And $Z_i = 1 + \delta_i$, where δ_i is the counterterms of fields. So for $i = 2$, i.e., the electron field

$$\begin{aligned} \delta_2 &= \frac{d}{d\bar{p}} \Sigma(p) \Big|_{p^2 = -M^2} = \frac{d}{d\bar{p}} \left(\text{---} \text{---} \text{---} \right) \\ &= \frac{-e^2}{16\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E \right) (1 - \epsilon \ln M) \\ &= \frac{-e^2}{16\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right), \end{aligned} \quad (2.121)$$

which leads to

$$\gamma_2 = \frac{e^2}{16\pi^2} \equiv \frac{\alpha}{4\pi}. \quad (2.122)$$

Also $i = 3$ is for the photon field

$$\begin{aligned} \delta_3 &= \Pi(p^2) \Big|_{p^2 = -M^2} = \left(\text{---} \text{---} \text{---} \right) = \left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2} \right) \Pi(p^2) \\ &= \frac{-e^2}{12\pi^2} \left(\frac{\Gamma(\epsilon)}{(M^2)^{\epsilon/2}} + O(1) \right) \simeq \frac{-e^2}{12\pi^2} \left(\frac{2}{\epsilon} - \gamma_E - 2 \ln M \right). \end{aligned} \quad (2.123)$$

Thus,

$$\gamma_3 = \frac{e^2}{12\pi^2} \equiv \frac{\alpha}{4\pi} \left(\frac{4}{3} \right). \quad (2.124)$$

For calculating the β -function, we first briefly review a very important solution in QED renormalization. Consider the Ward-Takahashi identity

$$(p - q)_\mu \Gamma^\mu(p, q) = S_F^{-1}(p) - S_F^{-1}(q), \quad (2.125)$$

when we take the limit that $p \rightarrow q$ and move it to the right, thus we have

$$\Gamma^\mu(p, p) = \frac{\partial S_F^{-1}(p)}{\partial p_\mu}. \quad (2.126)$$

Notice $S_F^{-1}(p) = \gamma^\mu p_\mu - m - \Sigma(p)$, thus we have

$$\Gamma^\mu(p, p) = \gamma^\mu - \frac{\partial \Sigma(p)}{\partial p_\mu}. \quad (2.127)$$

Beside the momentum p , the propagator can be renormalized as

$$S_F^{-1}(p) = Z_2^{-1}(\gamma \cdot p - m_{\text{ph}}) + \dots, \quad (2.128)$$

which means

$$\frac{\partial S_F^{-1}}{\partial p_\mu} \approx Z_2^{-1} \gamma^\mu. \quad (2.129)$$

And for the renormalized vertex, we have

$$\Gamma^\mu = Z_1^{-1} \Gamma_R^\mu. \quad (2.130)$$

In low energy approximation, the vertex just $\Gamma_R^\mu \rightarrow \gamma^\mu$. Thus, comparing (2.127) and (2.130), we have

$$Z_1 = Z_2. \quad (2.131)$$

Thus, $\delta_1 = \delta_2$. Then we immediately have

$$0 = \frac{\partial e_0}{\partial M} = \frac{\partial}{\partial M} \left[e \left(1 - \delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) \right] \quad (2.132)$$

$$= e \frac{\partial}{\partial M} \left(1 - \delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) + \left(1 - \delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) \frac{\partial e}{\partial M} \quad (2.133)$$

$$\simeq e \frac{\partial}{\partial M} \left(-\delta_1 + \delta_2 + \frac{1}{2} \delta_3 \right) + \frac{\partial e}{\partial M}. \quad (2.134)$$

Therefore, we get the final result for the β -function:

$$\begin{aligned} \beta(e) &= M \frac{\partial}{\partial M} e = -e M \frac{\partial}{\partial M} \left(\underbrace{-\delta_1 + \delta_2}_{\delta_1 = \delta_2} + \frac{1}{2} \delta_3 \right) \\ &= \frac{e^3}{12\pi^2} \equiv e \frac{\alpha}{4\pi} \left(\frac{4}{3} \right). \end{aligned} \quad (2.135)$$

4. Evolution of coupling

4.1 Solution of Callan-Symanzik equation

Now we would like to solve the solutions of Callan-Symanzik equation. We firstly start with the 2-point function $G^{(2)}(p)$, which generally can be written as:

$$G^{(2)}(p) = \frac{i}{p^2} f\left(\frac{-p^2}{M^2}\right), \quad (2.136)$$

where $f\left(\frac{-p^2}{M^2}\right)$ is a dimensionless function, which only depends on $\frac{-p^2}{M^2}$. In momentum space, without the amputation, which means we add the external lines to contribute the momentum p in the propagator. We have:

$$M \frac{\partial}{\partial M} G^{(2)}(p) = \frac{i}{p^2} \left(\frac{2p^2}{M^2}\right) f'\left(\frac{-p^2}{M^2}\right), \quad (2.137)$$

$$p \frac{\partial}{\partial p} G^{(2)}(p) = -2G^{(2)}(p) - \frac{i}{p^2} \left(\frac{2p^2}{M^2}\right) f'\left(\frac{-p^2}{M^2}\right). \quad (2.138)$$

Therefore, we have:

$$M \frac{\partial}{\partial M} G^{(2)}(p) = \left(-p \frac{\partial}{\partial p} - 2\right) G^{(2)}(p). \quad (2.139)$$

Putting this results into the Callan-Symanzik equation for 2-point function, we then have:

$$\left(p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda)\right) G^{(2)}(p^2, \lambda, M) = 0. \quad (2.140)$$

This is an equation for a current. We can firstly write the solution in free theory, i.e. $\lambda = 0$ which means $\beta = \gamma = 0$:

$$G^{(2)}(p) = \frac{i}{p^2} \bar{G}_0, \quad (2.141)$$

where $\bar{G}_0$ is a constant function in p^2 . But one can not happy if we only have the free theory. Therefore, we need to solve the renormalization group equation with a more general form. For that, we firstly introduce some new variables to make the PDE simpler:

$$t = \ln\left(\frac{p}{M}\right), \quad (2.142)$$

$$v(x) = -\beta(\lambda), \quad (2.143)$$

$$\rho(x) = 2\gamma(\lambda) - 2, \quad (2.144)$$

$$x = \lambda. \quad (2.145)$$

Now we call the new function as $D(t, x)$, which is nothing but the $G^{(2)}$. So (2.140) becomes

$$\left(\frac{\partial}{\partial t} + v(x) \frac{\partial}{\partial x} - \rho(x)\right) D(t, x) = 0. \quad (2.146)$$

The solution of $D(t, x)$ is somehow like a current with the current density

$$j(t, x) = \rho(t, x) v(t, x). \quad (2.147)$$

So the solution of a current can be expressed as

$$D(t, x) = D_0(\bar{x}(t, x)) \exp\left[\int_0^t dt' \rho(\bar{x}(t', x))\right], \quad (2.148)$$

with the boundary condition and the initial condition:

$$\frac{\partial}{\partial t'} \bar{x}(t', x) = -v(\bar{x}), \quad \bar{x}(0, x) = x, \quad (2.149)$$

where $\bar{x}(t, x)$ is the position of current volum element at t and x . So the Callan-Symanzik equation has the solution:

$$G^{(2)}(p^2, \lambda, M) = \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp \left[\int_{p'=M}^{p'=p} d \ln \left(\frac{p'}{M} \right) 2 [1 - \bar{\gamma}(\bar{\lambda}(p', \lambda))] \right]. \quad (2.150)$$

It is important to note that $\bar{G}_0(\bar{\lambda}(p, \lambda))$ is an unknown function, which does not directly corresponds to the propagator. To fix this function, we firstly put (2.136) into (2.140):

$$0 = \left(p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda) \right) \left[\frac{i}{p^2} f \left(\frac{p^2}{M^2}, \lambda \right) \right] \quad (2.151)$$

$$= \frac{i}{p^2} \left(-2 + p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} + 2 - 2\gamma(\lambda) \right) f \left(\frac{p^2}{M^2}, \lambda \right) \quad (2.152)$$

Thus, we have

$$\left(p \frac{\partial}{\partial p} - \beta \frac{\partial}{\partial \lambda} - 2\gamma(\lambda) \right) f \left(\frac{p^2}{M^2}, \lambda \right) = 0. \quad (2.153)$$

So the solution is similar with $G^{(2)}$:

$$f \left(\frac{p^2}{M^2}, \lambda \right) = \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp \left[-2 \int_M^p d \ln \left(\frac{p'}{M} \right) \gamma(\bar{\lambda}) \right]. \quad (2.154)$$

Therefore, we can take the another form of the solution by combine the solution of f and (2.136)

$$G^{(2)}(p^2, \lambda, M) = \frac{i}{p^2} \bar{G}_0(\bar{\lambda}(p, \lambda)) \exp \left[-2 \int_M^p d \ln \left(\frac{p'}{M} \right) \gamma(\bar{\lambda}(p', \lambda)) \right]. \quad (2.155)$$

Now, we can take the boundary $p = M$, which all the integration part will vanish. So we have

$$\bar{G}_0(\lambda) = -iM^2 \cdot G^{(2)}(M^2, \lambda, M). \quad (2.156)$$

We can then go to the specific case and use the perturbative expansion of propagator to get each order of $\bar{G}_0(\lambda)$.

4.2 Renormalization group equation

Now the boundary condition and the initial condition in (2.149) are changed to the so-called **renormalization group equation**:

$$\frac{d}{d \ln(p/M)} \bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}), \quad \bar{\lambda}(M, \lambda) = \lambda. \quad (2.157)$$

This equation shows that the flow of the running coupling constant $\bar{\lambda}(p)$ with the speed $\beta(\bar{\lambda})$. And we set the initial energy scale is M , and then set the value at this point is the original coupling constant λ . Now we expand $\bar{\lambda}$ to the first order and use (2.117):

$$\frac{d}{d \ln(p/M)} \bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}) = \frac{3}{16\pi^2} \bar{\lambda}^2 + O(\bar{\lambda}^3). \quad (2.158)$$

So we actually are solving the ODE

$$\frac{d\bar{\lambda}}{dt} = A\bar{\lambda}^2 \quad (2.159)$$

It is easy to solve it by integrating

$$\int_{\bar{\lambda}(M)=\lambda}^{\bar{\lambda}(p)} \bar{\lambda}'^{-2} d\bar{\lambda}' = \int_0^{\ln(p/M)} A dt' \quad (2.160)$$

Thus, we have

$$\frac{1}{\bar{\lambda}(p)} = \frac{1}{\bar{\lambda}} - A \ln(p/M) \quad (2.161)$$

$$= \frac{1 - A\bar{\lambda}(M) \ln(p/M)}{\bar{\lambda}(M)}. \quad (2.162)$$

Put all the parameters back, we have

$$\ln\left(\frac{p}{M}\right) = \left(\frac{3}{16\pi^2}\right)^{-1} \left(\frac{1}{\bar{\lambda}(M)} - \frac{1}{\bar{\lambda}(p)}\right). \quad (2.163)$$

Take $\bar{\lambda}(M) = \lambda$, which leads to the expression for the running coupling $\bar{\lambda}$:

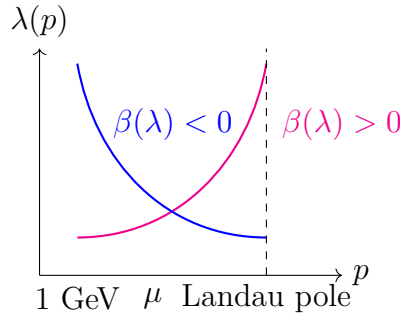
$$\bar{\lambda}(p) = \frac{\lambda}{1 - \frac{3}{16\pi^2} \lambda \ln(p/M)}. \quad (2.164)$$

So $\bar{\lambda}(p)$ grows for increasing scales p . And there is a pole when

$$1 - \frac{3}{16\pi^2} \lambda \ln\left(\frac{p}{M}\right) = 0, \quad (2.165)$$

which is called the **Landau pole**:

$$p = M \exp\left(\frac{16\pi^2}{3\lambda}\right). \quad (2.166)$$



For $p^2 \ll M^2$, theory becomes trivial:

$$\bar{\lambda}(p) \sim \frac{\bar{\lambda}(M)}{\lim_{x \rightarrow 0} (1 - \ln x)} \rightarrow 0_+ \quad (2.167)$$

which corresponds to the free theory.

Generally, we have different cases for the β function, and that corresponds to the different background theory

$$\frac{d}{d \ln(p/M)} \bar{\lambda}(p, \lambda) = \beta(\bar{\lambda}) \begin{cases} > 0 & \text{IR free } (\phi^4\text{-theory, QED, ...}) \\ = 0 & \text{conformal theory } (N = 4 \text{ SYM}) \\ < 0 & \text{UV free/asymptotically free (QCD, SU(N) gauge theory).} \end{cases} \quad (2.168)$$

4.3 Higgs mass and fate of universe

As the second example, we consider the Higgs mechanism. The story begin with the Lagrangian:

$$\mathcal{L}_{\text{unbroken}} = \bar{\psi}(i\gamma^\mu \partial_\mu)\psi + \frac{1}{2}(\partial^\mu \phi)(\partial_\mu \phi) - V(\phi) + y_\psi \phi \bar{\psi}\psi. \quad (2.169)$$

where $y_\psi \phi \bar{\psi}\psi$ is the Higgs-Yukawa interaction, and we dropped all the gauge terms. To get the spontaneous symmetry breaking, we take the potential as

$$V(\phi) = \mu^2 \phi^2 + \lambda \phi^4. \quad (2.170)$$

The spontaneous symmetry breaking happens when

$$\phi = H + v. \quad (2.171)$$

where H is the Higgs field, v is the vacuum expectation value, which in experiment measure are

$$v \simeq \frac{1}{\sqrt{\sqrt{2}G_F}} \simeq 246 \text{ GeV} \quad (2.172)$$

So plug (2.171) into the Lagrangian, we have the spontaneous symmetry breaking Lagrangian:

$$\mathcal{L}_{\text{broken}} = \bar{\psi}(i\partial)\psi + \frac{1}{2}(\partial_\mu H)(\partial^\mu H) - \underbrace{\frac{\lambda v^2}{\frac{1}{2}M_H^2}}_{\frac{1}{2}M_H^2} H^2 - \lambda v H^3 - \frac{\lambda}{4} H^4 - \underbrace{\frac{1}{\sqrt{2}} y_\psi v (1 + H/v)}_{m_\psi} \bar{\psi}\psi \quad (2.173)$$

This mechanism turns out to the particle mass by the Higgs interact with itself and the other fields. So the coupling constant renormalization is very similar with ϕ^4 theory and Yukawa theory:

$$-i\lambda + \underbrace{\text{loop diagrams}}_{O(\lambda^2)} + \underbrace{\text{box diagram}}_{O(y_\psi^4)} \quad (2.174)$$

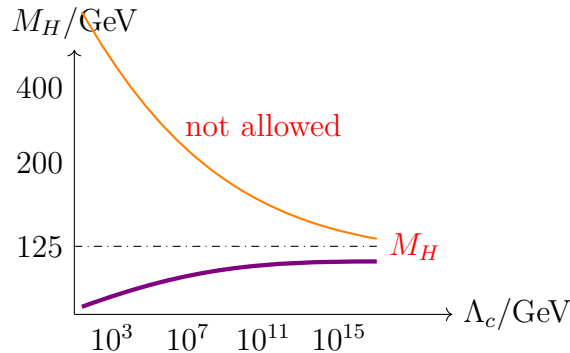
So the solution for $\lambda(Q)$ at scale $Q^2 \gg v$ is:

$$\lambda(Q) = \lambda(v) \frac{1}{1 - \frac{3}{16\pi^2} \lambda(v) \ln(Q/v)} \quad (2.175)$$

Cut off Λ_c corresponds to the Landau pole, up to which perturbative Higgs sector holds:

$$\Lambda_c = v \exp\left(\frac{16\pi^2}{3\lambda}\right) \simeq v \exp\left(\frac{16\pi^2 v^2}{M_H^2}\right) \quad (2.176)$$

where $M_H^2 = 2\lambda v^2$. So the area above the orange line are forbidden.



Now we consider fermion as the top quark, which is the heaviest fermion the standard model. It has contributions to $\beta(\lambda)$ from Higgs-Yukawa interaction, which the box diagram gives

$$Q \frac{d}{dQ} \lambda = \frac{1}{64\pi^2} (-3y_t^4 + 6\lambda y_t^2 + 12\lambda^2) + \dots \quad (2.177)$$

It is interesting that the first term is negative, which means it evolves in the opposite way with the Higgs. Solving this equation for small λ ($\lambda \ll 1$), we have

$$\lambda(Q) = \lambda(v) + \frac{1}{32\pi^2} \left(-\frac{3m_t^4}{v^4} \right) \ln \left(\frac{Q}{v} \right) \quad (2.178)$$

since $m_t = y_t \frac{v}{\sqrt{2}}$, $m_t \sim 172$ GeV, so in principle, $\lambda(Q)$ can be negative at large Q ! That means the vacuum will be super unstable. So to avoid that, we require a bound on Higgs mass:

$$M_H^2 \geq \frac{v^2}{16\pi^2} \left(-\frac{3m_t^4}{v^4} \right) \ln \left(\frac{Q}{v} \right), \quad (2.179)$$

such that we can keep $\lambda(Q) > 0$.

5. Renormalization for local operators and evolution of mass parameter

We first introduce the idea of local operator. The local operator is multiple field operator compositing in the same space-time position, namely the production of fields. This will happen when the energy scale is much lower than the high energy scattering, e.g. weak decay. So to understand this, we can recall the beta decay in electroweak interaction.

The electroweak interaction between left fermions and W-boson is:

$$\delta\mathcal{L} = \frac{g_{ew}}{\sqrt{2}} W_\mu \bar{\psi} \gamma^\mu (1 - \gamma_5) \psi. \quad (2.180)$$

And the propagator of W-boson is

$$\mu \text{ --- } W^- \text{ --- } \nu = \frac{-ig^{\mu\nu}}{q^2 - M_W^2 + i\epsilon} \quad (2.181)$$

In the low energy limit $q^2 \ll M_W^2$, the vertex and propagator of β -decay reduce to the 4-Fermi interaction vertex:

$$\begin{array}{ccc} \begin{array}{c} d \text{ --- } \swarrow \\ \nwarrow \text{ --- } \bar{u} \\ \text{--- } W^- \text{ ---} \\ \swarrow \text{ --- } e^- \\ \nwarrow \text{ --- } \bar{\nu}_e \end{array} & \xrightarrow{q^2 \ll M_W^2} & \begin{array}{c} d \text{ --- } \swarrow \\ \nwarrow \text{ --- } \bar{u} \\ \text{--- } \bullet \text{ ---} \\ \swarrow \text{ --- } e^- \\ \nwarrow \text{ --- } \bar{\nu}_e \end{array} \end{array} \quad (2.182)$$

So we can reduce a process propagate from μ to ν to the process happens in a fixed point due to the reaction happening so fast, i.e.:

$$\begin{array}{c} d \text{ --- } \swarrow \\ \nwarrow \text{ --- } \bar{u} \\ \text{--- } \bullet \text{ ---} \\ \swarrow \text{ --- } e^- \\ \nwarrow \text{ --- } \bar{\nu}_e \end{array} = \frac{g_{ew}^2}{2M_W^2} (\bar{\psi} \gamma^\mu (1 - \gamma_5) \psi)_{\text{quark}} (\bar{\psi} \gamma_\mu (1 - \gamma_5) \psi)_{\text{lep}} = \frac{g_{ew}^2}{2M_W^2} \mathcal{O}(x). \quad (2.183)$$

Generally, this special vertex can be seen as a Green's function:

$$\langle \psi(p_1) \bar{\psi}(p_2) \psi(p_3) \bar{\psi}(p_4) \mathcal{O}(0) \rangle = \text{diagram} \quad (2.184)$$

5.1 Renormalized Green's functions with local operator $\mathcal{O}$ in ϕ^4 -theory

Now we consider the renormalization of the local operator. Since the vertex is described by the Green's function, we can then consider a general form:

$$G^{(n,\mathcal{O})}(p_1 \dots p_n, k) = \langle \phi(p_1) \dots \phi(p_n) \mathcal{O}(k) \rangle \quad (2.185)$$

$$= Z^{-n/2}(\mu) \underbrace{Z_{\mathcal{O}}^{-1}(M)}_{\text{operator renormalization}} \underbrace{\langle \phi_0(p_1) \dots \phi_0(p_n) \rangle}_{\text{bare fields}} \underbrace{\mathcal{O}_0(k)}_{\text{bare operator}} \quad (2.186)$$

$$= Z^{-n/2}(\mu) Z_{\mathcal{O}}^{-1}(M) G_0^{(n,\mathcal{O})}(p_1 \dots p_n, k) \quad (2.187)$$

where $\mathcal{O} = \phi^2$ or ϕ^4 or $\dots$. The lower index "0" represent the bare quantity, and M is the renormalization scale. The bare operator and the renormalized operator have the relation:

$$\mathcal{O}_{\text{bare}} = Z_{\mathcal{O}}(M) \mathcal{O}_{\text{ren}}. \quad (2.188)$$

So we can derive the Callan-Symanzik equation by using

$$0 = M \frac{d}{dM} G_0^{(n,\mathcal{O}_0)} \quad (2.189)$$

$$= M \frac{d}{dM} (Z^{n/2} Z_{\mathcal{O}} G^{(n,\mathcal{O})}) \quad (2.190)$$

$$= \left(M \frac{\partial}{\partial M} Z^{n/2} \right) Z_{\mathcal{O}} G^{(n,\mathcal{O})} + Z^{n/2} \left(M \frac{\partial}{\partial M} Z_{\mathcal{O}} \right) G^{(n,\mathcal{O})} + Z^{n/2} Z_{\mathcal{O}} \left(M \frac{\partial}{\partial M} G^{(n,\mathcal{O})} \right), \quad (2.191)$$

dividing $Z^{n/2} Z_{\mathcal{O}}$ on both side, we have

$$0 = \left(\frac{M}{Z^{n/2}} \frac{\partial Z^{n/2}}{\partial M} \right) G^{(n,\mathcal{O})} + \left(\frac{M}{Z_{\mathcal{O}}} \frac{\partial Z_{\mathcal{O}}}{\partial M} \right) G^{(n,\mathcal{O})} + M \frac{\partial}{\partial M} G^{(n,\mathcal{O})} \quad (2.192)$$

$$= \left(M \frac{\partial}{\partial M} \ln Z^{n/2} \right) G^{(n,\mathcal{O})} + \left(M \frac{\partial}{\partial M} \ln Z_{\mathcal{O}} \right) G^{(n,\mathcal{O})} + M \frac{\partial}{\partial M} G^{(n,\mathcal{O})}, \quad (2.193)$$

where we can define the anomalous dimension respectively for field and local operator as:

$$\gamma = \frac{M}{2} \frac{\partial}{\partial M} \ln Z_{\lambda}(M) \quad (2.194)$$

$$\gamma_{\mathcal{O}} = M \frac{\partial}{\partial M} \ln Z_{\mathcal{O}}(M). \quad (2.195)$$

Then we have the Callan-Symanzik equation:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \gamma_{\mathcal{O}}(\lambda) \right) G^{(n,\mathcal{O})} = 0, \quad (2.196)$$

Now we compute $\gamma_{\mathcal{O}}$ for N-point Green's function. The most general case can be expressed as

$$\text{Diagram with red dot and cross} + \sum_i \left(\text{Diagram with red dot and circle} + \text{Diagram with red dot and cross} \right) + \text{Diagram with red cross}, \quad (2.197)$$

where the red point and red cross represent the operator vertex and the operator counterterm.

5.2 Renormalization for local operators in massless ϕ^4 -theory

As an example to help us understanding this. We consider $\mathcal{O} = \phi^2$ inserted in two-point function:

$$\text{Diagram with red dot and cross} = \langle \Omega | \phi(q) \phi(p) \phi^2(k) | \Omega \rangle = G^{(2, \phi^2)}. \quad (2.198)$$

Thus, we can expand it as the normal Dyson series:

$$G^{(2, \phi^2)}(q, p; k) = \sum_{m=0}^{\infty} \frac{i^m}{m!} \int \left[\prod_{a=1}^m d^d x_a \right] \left(-\frac{\lambda}{4!} \right)^m \langle 0 | T \{ \phi(q) \phi(p) \phi^2(k) [\phi^4(x_1) \dots \phi^4(x_m)] \} | 0 \rangle_{\text{connect}}. \quad (2.199)$$

The leading order, namely the tree-level non-amputated Green's function is just simply the contraction of the field and the operator:

$$G_{\text{tree}}^{(2, \phi^2)}(q, p; k) = \text{Diagram with red dot and cross} = \int d^d x e^{-ik \cdot x} \langle 0 | T \{ \phi(q) \phi(p) \phi_A(x) \phi_B(x) \} | 0 \rangle \quad (2.200)$$

$$= \int d^d x e^{-ik \cdot x} \left[\overbrace{\phi(q) \phi(p) \phi_A(x) \phi_B(x)} + \overbrace{\phi(q) \phi(p) \phi_A(x) \phi_B(x)} \right] \quad (2.201)$$

$$= \left(\frac{i}{q^2} \right) \left(\frac{i}{p^2} \right) \cdot 2. \quad (2.202)$$

To consider the renormalization for the operator, we need to compute 1-PI contribution at 1-loop (non-amputated):

$$G_{1\text{-loop}}^{(2,\phi^2)}(q, p; k) = \text{u+v} \begin{array}{c} \text{diagram: a circle with an incoming arrow from the top labeled } ik, \text{ and two outgoing arrows from the bottom labeled } q \text{ and } p. \end{array} = \int d^d x d^d z e^{-ik \cdot x} \left\langle 0 \left| T \left\{ \phi(q) \phi(p) \left[\frac{-i\lambda}{4!} \phi^4(z) \right] \left[\frac{1}{2} \phi_A(x) \phi_B(x) \right] \right\} \right| 0 \right\rangle \quad (2.203)$$

$$= \frac{-i\lambda}{2} \int d^d x d^d z e^{-ik \cdot x} \left[\overbrace{\phi(q) \phi(p)} \quad \overbrace{\phi(z) \phi(z) \phi(z) \phi(z)} \quad \overbrace{\phi_A(x) \phi_B(x)} \right] \quad (2.204)$$

$$= \left(\frac{i}{q^2} \right) \left(\frac{i}{p^2} \right) \int \frac{d^d v}{(2\pi)^d} (-i\lambda) \frac{i}{v^2} \frac{i}{(k+v)^2}, \quad (2.205)$$

with some calculation, one has:

$$G_{1\text{-loop}}^{(2,\phi^2)}(q, p; k) = \frac{i}{p^2} \frac{i}{q^2} \left(-\frac{\lambda}{(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{\Delta^{2-d/2}} \right) \quad (2.206)$$

with the counterterm:

$$\text{diagram: a vertex with two outgoing arrows labeled } q \text{ and } p, \text{ crossed out with a red circle and an } \times. = \frac{i}{p^2} \frac{i}{q^2} 2\delta\phi^2 \quad (2.207)$$

Comparing of coefficients in these two graphs, we then can find the counterterm at scale $p^2 = -M^2$ should be:

$$\delta\phi^2 = \frac{\lambda}{2(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{(M^2)^{2-d/2}}. \quad (2.208)$$

This implies the operator renormalization is

$$Z_{\phi^2} = 1 - \delta\phi^2 = 1 - \frac{\lambda}{2(4\pi)^{d/2}} \frac{\Gamma(2 - \frac{d}{2})}{(M^2)^{2-d/2}} \quad (2.209)$$

$$\stackrel{d=4-\epsilon}{=} 1 - \frac{\lambda}{(4\pi)^2} \frac{1}{\epsilon} + \text{finite terms}. \quad (2.210)$$

Using $\ln(1+x) \approx x$, we can obtain the anomalous dimension of the operator in $d = 4 - \epsilon$:

$$\gamma_{\phi^2} = M \frac{\partial}{\partial M} \ln Z_{\phi^2} = \frac{-\lambda}{2(4\pi)^{d/2}} \Gamma\left(2 - \frac{d}{2}\right) \cdot \left[-(4-d) M^{-(4-d)} \right] \quad (2.211)$$

$$\stackrel{d=4-\epsilon}{=} \frac{-\lambda}{2(4\pi)^{2-\epsilon/2}} \Gamma\left(\frac{\epsilon}{2}\right) \cdot \left[-\epsilon M^{-\epsilon} \right] \quad (2.212)$$

$$= \frac{\lambda}{16\pi^2}. \quad (2.213)$$

More generally, if we insert the operator into an n-point Green's function, we then need to consider the field renormalization with the operator renormalization $Z_\phi^{n/2} Z_{\mathcal{O}}$, therefore the anomalous dimension becomes:

$$\gamma_{\mathcal{O}}(\lambda) = M \frac{\partial}{\partial M} \left(-\delta_{\mathcal{O}} + \frac{n}{2} \delta_Z \right), \quad (2.214)$$

with the approximation:

$$\ln(Z_\phi^{n/2} Z_\mathcal{O}) = \frac{n}{2} \ln Z_\phi + \ln Z_\mathcal{O} \approx \frac{n}{2} \delta_Z - \delta_\mathcal{O}. \quad (2.215)$$

This case can be treated as renormalized ϕ^4 -theory with effective operator $-\frac{1}{2}m^2\phi^2$, where ϕ^2 is local operator inserted in Green's function. So the Callan-Symanzik equation with mass anomalous dimension $m(\mu)$ is

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \gamma_{\phi^2} m^2 \frac{\partial}{\partial m^2} \right) G^{(n, \phi^2)}(\{p_i\}, M, \lambda, m) = 0 \quad (2.216)$$

5.3 Evolution of mass parameter

Moreover, one can write the generalization to local operators $\mathcal{O}_i$ in ϕ^4 -theory as:

$$\mathcal{L}(C_i) = \mathcal{L}_{\text{eff}} + \sum_i C_i \mathcal{O}_i(x) \quad (2.217)$$

where C_i is the coupling of local operators. So the Callan-Symanzik equation is

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \gamma_i(\lambda) C_i \frac{\partial}{\partial C_i} \right) G^{(n)}(M, \lambda, \{C_i\}) = 0 \quad (2.218)$$

To guarantee the dimension of Lagrangian is $[\text{Mass}]^4$ in 4-dimension spacetime, so we need to explicit mass dimension in coupling C_i as:

$$C_i = \rho_i M^{4-d_i} \quad (2.219)$$

where ρ_i is dimensionless, d_i is dimension of $\mathcal{O}_i$. Therefore, one can see that the variables change to $(M, \lambda, \{\rho_i\})$. To make the partial derivative independent to ρ_i , we have

$$M \frac{\partial}{\partial M} \Big|_{\rho_i} = M \frac{\partial}{\partial M} \Big|_{C_i} + \sum_i \left(M \frac{\partial C_i}{\partial M} \Big|_{\rho_i} \right) \frac{\partial}{\partial C_i} \quad (2.220)$$

Notice that

$$\rho_i \frac{\partial}{\partial \rho_i} = \rho_i M^{4-d_i} \frac{\partial}{\partial C_i} = C_i \frac{\partial}{\partial C_i} \quad (2.221)$$

so we have

$$M \frac{\partial}{\partial M} \Big|_{C_i} = M \frac{\partial}{\partial M} \Big|_{\rho_i} - \sum_i (4 - d_i) \rho_i \frac{\partial}{\partial \rho_i} \quad (2.222)$$

Then the Callan-Symanzik equation becomes:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i (\gamma_i(\lambda) + d_i - 4) \rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0, \quad (2.223)$$

where we can define the β function for new coupling ρ_i :

$$\beta_i(\lambda) = (\gamma_i(\lambda) + d_i - 4) \rho_i, \quad (2.224)$$

then the Callan-Symanzik equation can be reduced as:

$$\left(M \frac{\partial}{\partial M} + \beta(\lambda) \frac{\partial}{\partial \lambda} + n\gamma(\lambda) + \sum_i \beta_i \rho_i \frac{\partial}{\partial \rho_i} \right) G^{(n)} = 0. \quad (2.225)$$

With this, we can easily solve the running coupling ρ_i from the renormalization group equation for β_i

$$\frac{d}{d \ln(p/M)} \bar{\rho}_i = \beta_i(\bar{\rho}_i, \lambda). \quad (2.226)$$

Since β_i in the low energy limit, namely $\gamma_i(\lambda) \ll 1$, is dominated by the classical dimension $d_i - 4$, we then have

$$\frac{d}{d \ln(p/M)} \bar{\rho}_i \simeq (d_i - 4 + \dots) \cdot \bar{\rho}_i \quad (2.227)$$

Therefore, the solution in low energy limit reads:

$$\bar{\rho}_i = \rho \left(\frac{p}{M} \right)^{d_i - 4}. \quad (2.228)$$

Now we can clearly see that for the operators with mass dimension $d_i > 4$ will vanish as $p \rightarrow 0$, e.g., ϕ^6 theory with $d_i = 6$. These operators are called the **irrelevant operators**, which usually decrease fast until vanish in low energy limit, namely one can never feel these operators' contribution. On the contrary, the **relevant operators** are the operators with mass dimension $d_i < 4$, e.g., the mass term $\frac{1}{2}m^2\phi^2$, which will exponentially increase in low energy limit. And the operators like ϕ^4 are called the **marginal operators**, which do not changing with the energy scale due to $d_i = 4$. This type operators are only determined by the anomalous dimension $\gamma_i(\lambda)$.

This reminds us that the effective field theory truly works even though we don't know all the details in high energy level. But we still can derive the precise renormalizable Lagrangian in low energy level.